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EUV TRANSMISSIVE MEMBRANE, PELLICLE, AND EXPOSURE METHOD

Abstract

Provided is an EUV transmissive membrane having a three-layer configuration composed of a metallic beryllium layer (**12**) having a first side and a second side, a first nitride layer (**14***a*) that covers the first side (**12***a*) of the beryllium layer (**12**), wherein the first nitride layer includes at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride, and a second nitride layer (**14***b*) that covers the second side (**12***b*) of the beryllium layer (**12**), wherein the second nitride layer includes at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride. The EUV transmissive membrane (**10**) has an EUV transmittance of 88% or more at a wavelength of 13.5 nm.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of PCT/JP2024/007655 filed Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present disclosure relates to an EUV transmissive membrane, pellicle, and exposure method.
2. Description of the Related Art
[0003] With miniaturization in semiconductor manufacturing process advancing year by year, various improvements have been made in each step. Particularly, in a photolithography step, extreme ultraviolet (EUV) light having a wavelength of 13.5 nm has begun to be used in place of conventional ArF exposure having a wavelength of 193 nm. As a result, the wavelength was reduced to 1/10 or less at once, and optical properties thereof were completely different. Since there is no material having a high transmittance to EUV light, however, there is still no practical pellicle, which serves as a particle adhesion-preventing membrane of, for example, a photomask (reticle). For this reason, device manufacturers currently cannot use pellicles when manufacturing semiconductor devices.
[0004] Therefore, a pellicle membrane has been developed. It is desirable to use Si, Be, Y, Zr, or the like having a high EUV transmittance as a core material of the pellicle membrane. Patent Literature 1 (JP6858817B) discloses a pellicle membrane including a core layer that contains a material substantially transparent for EUV radiation such as (poly-)Si and a cap layer that contains a material absorbing IR radiation.
[0005] In addition, Patent Literature 2 (JP2020-98227A) discloses a pellicle membrane stretched on one end face of a pellicle frame, the pellicle membrane having a main layer of single-crystal Si, and graphene on one side or both sides of the main layer. It has been proposed that when the main layer has graphene, the pellicle membrane is not damaged during pellicle fabrication and can exhibit sufficient mechanical strength.
CITATION LIST
Patent Literature
[0006] Patent Literature 1: JP6858817B [0007] Patent Literature 2: JP2020-98227A
SUMMARY OF THE INVENTION
[0008] The core material having high EUV transmittance used for the pellicle membrane as described above forms a natural oxide membrane of several nm on the surface thereof in air, and the oxide membrane absorbs EUV, resulting in a decrease in EUV transmittance of the pellicle membrane. In particular, Be is a material having high EUV transmittance, but a natural oxide membrane of 2 to 3 nm is considered to be formed on the surface thereof. The transmittance loss due to the oxide membrane is as large as 6 to 9%. In addition, the process of fabricating a pellicle membrane sometimes involves treatment using a gas other than air, such as fluorine or chlorine, an acid, or an alkali solution, which may generate a side reaction membrane on the surface of the core material, thereby decreasing the EUV transmittance. Therefore, it is desirable to form a protective layer on the surface of the core material to suppress the formation of these membranes.
[0009] However, while applying a reaction-inhibiting protective layer on the surface of a core material such as Si and Be can prevent a significant decrease in EUV transmittance of the pellicle

membrane, such a protective layer also has lower EUV transmittance than the pure core material, leading to a decrease in the EUV transmittance of the pellicle membrane as a whole. For example, Ru may be used as a protection layer for a Be core material. In the case of a pellicle membrane with a three-layer structure (Ru/Be/Ru) in which 1 to 2 nm of Ru is provided on the front and back surfaces of the Be core material, the Ru protective layer causes an EUV transmission loss of about 3 to 6%, which significantly degrades the performance of the pellicle membrane.
[0010] The inventors have recently found that by adopting a three-layer configuration of nitride layer/metallic beryllium layer/nitride layer, it is possible to provide an EUV transmissive membrane that exhibits high EUV transmittance.
[0011] Accordingly, an object of the present invention is to provide an EUV transmissive membrane or pellicle that exhibits high EUV transmittance. Another object of the present invention is to provide an exposure method using an EUV transmissive membrane.
[0012] The present disclosure provides the following aspects.
Aspect 1
[0013] An EUV transmissive membrane with a three-layer configuration consisting of: [0014] a metallic beryllium layer having a first side and a second side, [0015] a first nitride layer that covers the first side of the beryllium layer, wherein the first nitride layer comprises at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride, and [0016] a second nitride layer that covers the second side of the beryllium layer, wherein the second nitride layer comprises at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride, [0017] wherein the EUV transmissive membrane has an EUV transmittance of 88% or more at a wavelength of 13.5 nm.
Aspect 2
[0018] The EUV transmissive membrane according to aspect 1, wherein the EUV transmissive membrane has a thickness of 7 to 30 nm.
Aspect 3
[0019] The EUV transmissive membrane according to aspect 1 or 2, wherein the beryllium layer has a thickness of 5 to 25 nm.
Aspect 4
[0020] The EUV transmissive membrane according to any one of aspects 1 to 3, wherein the first nitride layer and the second nitride layer each have a thickness of 1 to 5 nm.
Aspect 5
[0021] A pellicle comprising: [0022] a substrate having a first side and a second side, and a cavity at a center of the substrate, [0023] an amorphous carbon layer that covers the first side of the substrate, and [0024] the EUV transmissive membrane according to any one of aspects 1 to 4, which covers a surface of the amorphous carbon layer opposite to the substrate, wherein the EUV transmissive membrane is exposed as a free-standing membrane constituting a bottom surface of the cavity at a same level as a surface of the amorphous carbon layer.
Aspect 6
[0025] The pellicle according to aspect 5, wherein the amorphous carbon layer has a thickness of 1 to 15 nm.
Aspect 7
[0026] The pellicle according to aspect 5 or 6, wherein the substrate is a Si substrate.
Aspect 8
[0027] The pellicle according to any one of aspects 5 to 7, further comprising a mask layer that covers the second side of the substrate.
Aspect 9
[0028] The pellicle according to aspect 8, wherein the mask layer is a SiO.sub.2 layer.
Aspect 10
[0029] An exposure method, comprising the steps of: [0030] providing a composite membrane with

a five-layer configuration, wherein the composite membrane comprises a protective layer containing amorphous carbon on both sides of the EUV transmissive membrane according to any one of aspects 1 to 4; [0031] installing the composite membrane with a five-layer configuration into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated; [0032] bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the composite membrane with a five-layer configuration to remove the protective layer; and [0033] installing the EUV transmissive membrane with a three-layer configuration, from which the protective layer has been removed, into an EUV exposure apparatus, then transmitting EUV through the EUV transmissive membrane to perform pattern exposure on a photosensitive substrate plate in the EUV exposure apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] FIG. **1** is a schematic cross-sectional view illustrating an embodiment of an EUV transmissive membrane according to the present invention.

[0035] FIG. **2**A is a process flow diagram illustrating the first half of a manufacturing procedure for an EUV transmissive membrane.

[0036] FIG. **2**B is a process flow diagram illustrating the second half of a manufacturing procedure for an EUV transmissive membrane.

DETAILED DESCRIPTION OF THE INVENTION

EUV Transmissive Membrane

[0037] FIG. **1** illustrates a schematic cross-sectional view of an EUV transmissive membrane **10** according to an embodiment of the present invention. The EUV transmissive membrane **10** is a membrane with a three-layer configuration consisting of a metallic beryllium layer **12**, a first nitride layer **14***a*, and a second nitride layer **14***b*. The metallic beryllium layer **12** has a first side **12***a* and a second side **12***b*. The first nitride layer **14***a* is a layer that covers the first side **12***a* of the metallic beryllium layer **12** and contains at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride. The second nitride layer **14***b* is a layer that covers the second side **12***b* of the metallic beryllium layer **12** and contains at least one selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride. The EUV transmissive membrane **10** has an EUV transmittance of 88% or more at a wavelength of 13.5 nm. Thus, by adopting a three-layer configuration of first nitride layer **14***a*/metallic beryllium layer **12**/second nitride layer **14***b*, it is possible to provide the EUV transmissive membrane **10** that exhibits high EUV transmittance.

[0038] As mentioned above, a core material having high EUV transmittance may form a natural oxide membrane of several nm on the surface thereof in air. In addition, the process of fabricating a pellicle membrane sometimes involves treatment using a gas other than air, such as fluorine or chlorine, an acid, or an alkali solution, which may generate a side reaction membrane on the surface of the core material. The formation of such a membrane decreases the EUV transmittance of the pellicle membrane. Therefore, it is desirable to form a protective layer on the surface of the core material to suppress the formation of these membranes. However, applying a reaction-inhibiting protective layer to the surface of the core material leads to a decrease in the EUV transmittance of the pellicle membrane as a whole. A possible approach to address this problem is to provide an amorphous carbon layer as a protective membrane that can be removed by bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with both sides of the EUV transmissive membrane. The above-described natural oxide membrane or side reaction membrane is formed before the pellicle membrane is fabricated, mounted on the EUV exposure apparatus, and subjected to the exposure process. However, the formation of the above-described membrane can be prevented by providing the amorphous carbon layer as a protective layer on both sides of the EUV transmissive membrane **10**. In particular, the amorphous carbon layer is effective in protecting the EUV transmissive membrane **10** from various agents (e.g., fluorine-based etching agent with strong reactivity) used in a pellicle membrane fabricating process (e.g., free-standing membrane forming step). When the amorphous carbon layer is provided directly on the metallic beryllium layer **12**, the amorphous carbon and metallic beryllium may react to form beryllium carbide. However, the EUV transmissive membrane **10** is composed of three layers of first nitride layer **14***a*/metallic beryllium layer **12**/second nitride layer **14***b* and provided with the amorphous carbon layer on the nitride layers **14***a* and **14***b*. Thus, an undesirable reaction between amorphous carbon and metallic beryllium can be prevented. On the other hand, after serving as a protective layer in the pellicle membrane fabricating process (e.g., free-standing membrane forming step), the amorphous carbon layer becomes an unnecessary membrane in terms of reducing the EUV transmittance. For this reason, it is possible to attempt to improve the EUV transmittance by bringing the amorphous carbon layer into contact with hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals for removal. The EUV transmissive membrane **10** thus obtained has a three-layer configuration consisting of the metallic beryllium layer **12**, the first nitride layer **14***a*, and the second nitride layer **14***b*, and no longer has a protective layer (amorphous carbon layer), so that a decrease in EUV transmittance due to the protective layer can be avoided. As a result, the EUV transmissive membrane **10** can maximize the high transmittance intrinsic to the three-layer configuration consisting of the metallic beryllium layer **12**, the first nitride layer **14***a*, and the second nitride layer **14***b*, that is, can exhibit high EUV transmittance during exposure.

[0039] As described above, the EUV transmissive membrane **10** has high EUV transmittance. The EUV transmissive membrane **10** has an EUV transmittance of 88% or more, preferably 92% or more, more preferably 93% or more, even more preferably 94% or more, particularly preferably 95% or more, and most preferably 96% or more at a wavelength of 13.5 nm. A higher EUV transmittance of the EUV transmissive membrane **10** is desirable, and the upper limit thereof is not particularly limited. Although ideally 100%, the EUV transmissive membrane **10** typically has a EUV transmittance of 99% or less, and more typically 98% or less.

[0040] The metallic beryllium layer **12** is a layer containing metallic beryllium as a main component. Here, the “main component” in the metallic beryllium layer **12** means a component that accounts for 50 mol % or more, preferably 70 mol % or more, more preferably 80 mol % or more, and even more preferably 90 mol % or more of the metallic beryllium layer **12**. The metallic beryllium layer **12** may also contain impurities in addition to metal beryllium as a main component. Accordingly, the metallic beryllium layer **12** may be composed of metallic beryllium and inevitable impurities. The thickness of the metallic beryllium layer **12** is preferably 5 to 25 nm, more preferably 7 to 20 nm, and even more preferably 9 to 15 nm.

[0041] The first nitride layer **14***a* and the second nitride layer **14***b* each contain at least one nitride selected from the group consisting of silicon nitride, beryllium nitride, boron nitride, and zirconium nitride. A particularly preferred nitride is silicon nitride. The advantages of providing the first nitride layer **14***a* and the second nitride layer **14***b* on both sides of the metallic beryllium layer **12** are as described above. The terms “silicon nitride”, “beryllium nitride”, “boron nitride”, and “zirconium nitride” as used herein each mean a comprehensive composition that allows not only a stoichiometric composition such as Si.sub.3N.sub.4, Be.sub.3N.sub.2, BN, and ZrN, but also a non-stoichiometric composition such as Si.sub.3N.sub.4-x, wherein 0<x<4, Be.sub.3N.sub.2-x, wherein 0<x<2, BN.sub.x, wherein 0<x<1, and ZrN.sub.x, wherein 0<x<1. The first nitride layer **14***a* and the second nitride layer **14***b* each preferably have a thickness of 1 to 5 nm, and more preferably 1 to 3 nm.

[0042] The first nitride layer **14***a*, the metallic beryllium layer **12**, and the second nitride layer **14***b* constitute, by only these layers, the EUV transmissive membrane **10**. The these layers contribute to the realization of high EUV transmittance while ensuring basic functions as a pellicle membrane (such as a particle adhesion preventing function). The thickness of the EUV transmissive membrane **10** is preferably 7 to 30 nm, more preferably 9 to 26 nm, and even more preferably 11 to 21 nm.

[0043] When the first nitride layer **14***a* and/or the second nitride layer **14***b* each contain beryllium nitride, the EUV transmissive membrane **10** preferably has a nitrogen concentration gradient region where nitrogen concentration decreases as closer to the metallic beryllium layer **12**. In other words, when the composition of beryllium nitride may include from the stoichiometric composition such as Be.sub.3N.sub.2 to the non-stoichiometric composition such as Be.sub.3N.sub.2-x, wherein $0<x<2$, as described above, the beryllium nitride constituting the beryllium nitride layer preferably has a gradient composition that is richer in beryllium as closer to the metallic beryllium layer **12**. As a result, it is possible to improve the adhesion between the nitride layers **14***a* and **14***b* (i.e., beryllium nitride layer) constituting the EUV transmissive membrane **10** and the metallic beryllium layer **12** and relieve the generation of stress caused by the difference in thermal expansion between these layers. That is, it is possible to improve the adhesion between these layers to suppress delamination, and to make delamination difficult as a thermal expansion relaxation layer between these layers in the case of absorbing EUV light and becoming a high temperature. The thickness of the nitrogen concentration gradient region is preferably smaller than that of each of the nitride layers **14***a* and **14***b*. In other words, the entire thickness of each of the nitride layers **14***a* and **14***b* does not need to be in the nitrogen concentration gradient region. For example, it is preferable that only a part of the thickness of each of the nitride layers **14***a* and **14***b*, for example, a region of preferably 10 to 70% and more preferably 15 to 50% of the thickness of each of the nitride layers **14***a* and **14***b* is the nitrogen concentration gradient region.

[0044] In the EUV transmissive membrane **10**, the main region for transmitting EUV is preferably in a form of the free-standing membrane. In other words, it is preferable that the substrate **20** or the like (e.g., Si substrate) used at the time of deposition remains as a border only at the outer edge of the EUV transmissive membrane **10** as a pellicle **11** illustrated in FIG. **2**B(*k*) described later, that is, no substrate or the like (e.g., Si substrate) remains in the main region other than the outer edge. In short, the main region preferably consists only of three layers of the first nitride layer **14***a*, the metallic beryllium layer **12**, and the second nitride layer **14***b*.

Manufacturing Method

[0045] After a composite membrane with a five-layer configuration including a protective layer that contains amorphous carbon on both sides of an EUV transmissive membrane is formed on a Si substrate, the EUV transmissive membrane or pellicle according to the present invention can be fabricated by removing an unnecessary portion of the Si substrate and the protective layer through etching to form a free-standing membrane. Accordingly, the main portion of the EUV transmissive membrane is in the form of the free-standing membrane in which no Si substrate remains as described above.

(1) Preparation of Si Substrate

[0046] First, as illustrated in FIG. **2**B, a Si substrate **28** for forming a composite membrane thereon is prepared. After the composite membrane composed of the second protective layer **16***b*, the second nitride layer **14***b*, the metallic beryllium layer **12**, the first nitride layer **14***a*, and the first protective layer **16***a* is formed on the Si substrate **28**, the main region (i.e., a region to be a free-standing membrane) other than the outer edge of the Si substrate is removed by etching. Accordingly, it is desirable to reduce the thickness of the Si substrate in the region to be formed into the free-standing membrane in advance in order to perform the etching efficiently in a short time. Therefore, it is desirable that a mask corresponding to the EUV transmission shape is formed on the Si substrate by employing a normal semiconductor process, and the Si substrate is etched by wet etching to reduce the thickness of the main region of the Si substrate to a predetermined thickness. The wet-etched Si substrate is cleaned and dried to prepare a Si substrate having a cavity formed by wet etching. The wet etching mask may be made of any material that is corrosion resistance to Si wet etchant, for example, SiO_2 is suitable for use. In addition, the wet etchant is not particularly limited as long as it is capable of etching Si. For example, TMAH (tetramethylammonium hydroxide) is preferred because it can be used under appropriate conditions, and very good anisotropic etching can be performed on Si.

(2) Formation of Composite Membrane

[0047] The composite membrane composed of the second protective layer **16***b*, the second nitride layer **14***b*, the metallic beryllium layer **12**, the first nitride layer **14***a*, and the first protective layer **16***a* is formed in order on the Si substrate. The composite membrane may be formed by any deposition method. An example of the preferred deposition method is the sputtering method. It is preferable that the metallic beryllium layer **12** is fabricated by sputtering using a pure Be target.

[0048] The first nitride layer **14***a* and the second nitride layer **14***b* are also preferably fabricated by sputtering. For example, the nitride layers **14***a* and **14***b* may be formed by (i) forming a Si, Be, B or Zr membrane by sputtering using a Si, Be, B, or Zr target and then irradiating the membrane with nitrogen plasma to cause a nitriding reaction of Si, Be, B, or Zr. The first nitride layer **14***a* and the second nitride layer **14***b* may also be fabricated by (ii) reactive sputtering. The reactive sputtering can be performed, for example, by adding nitrogen gas to the chamber during sputtering using a Si, Be, B, or Zr target, whereby Si, Be, B, or Zr and nitrogen react to each other to generate silicon nitride, beryllium nitride, boron nitride, or zirconium nitride. Alternatively, the nitride layers **14***a* and **14***b* may be directly formed by (iii) sputtering using a Si_3N_4, Be_3N_2, BN, or ZrN target. Nitrogen gas may be added to the chamber during the sputtering, whereby Si, Be, B, or Zr and nitrogen react to each other as in the case of the reactive sputtering (ii) above to promote the generation of silicon nitride, beryllium nitride, boron nitride, or zirconium nitride.

[0049] Each of the first protective layer **16***a* and the second protective layer **16***b* is a layer containing amorphous carbon. The layer containing amorphous carbon is advantageous in that, in addition to exhibiting high protection performance against various agents (e.g., fluorine-based etching agent with strong reactivity) used in a pellicle membrane fabricating process (e.g., free-standing membrane forming step), the layer reduces the impact of the residue on the EUV transmissive membrane **10** when the protective layers **16***a* and **16***b* are removed, and that the protective layers **16***a* and **16***b* are easily removed. The first protective layer **16***a* and the second protective layer **16***b* preferably contains amorphous carbon as a main component, and more preferably is composed of amorphous carbon. In general, amorphous carbon does not have a completely disordered atomic arrangement but microscopically has a crystal structure (i.e., it has crystallites). Such crystallites are often arranged in a disordered manner, making them amorphous as a whole. In particular, diamond-like carbon (DLC) refers to a material containing many crystallites with a cubic four-coordinated structure, such as diamond, while graphite-like carbon (GLC) refers to a material containing many crystallites with a planar three-coordinated structure, such as graphite. The first protective layer **16***a* and the second protective layer **16***b* may be carbon with a completely disordered atomic arrangement, carbon having crystallites in a disordered manner, or even amorphous carbon that is not completely dense but rather contains fine pores. Here, the “main component” in the first protective layer **16***a* and the second protective layer **16***b* means a component that accounts for 50% by weight or more, preferably 60% by weight or more, more preferably 70% by weight or more, and even more preferably 80% by weight or more of the total weight of the first protective layer **16***a* or the second protective layer **16***b*. Meanwhile, the first protective layer **16***a* and the second protective layer **16***b* may be composed only of amorphous carbon. The first protective layer **16***a* and the second protective layer **16***b* each preferably have a thickness of 1 to 15 nm, more preferably 2 to 12 nm, and even more preferably 3 to 10 nm. It is preferable that the formation of the amorphous carbon layer as the first protective layer **16***a* and the second protective layer **16***b* is performed by sputtering using a graphite target.

[0050] The method of forming the metallic beryllium layer **12**, the nitride layers **14***a* and **14***b*, and the protective layers **16***a* and **16***b* is not limited thereto. The metallic beryllium layer **12**, the nitride layers **14***a* and **14***b*, and the protective layers **16***a* and **16***b* may be formed in a one-chamber sputtering apparatus as in Examples described later, or a multi-chamber sputtering apparatus may be used to form the metallic beryllium layer **12**, the nitride layers **14***a* and **14***b*, and the protective layers **16***a* and **16***b* in separate chambers.

[0051] In the case where the first nitride layer **14***a* and the second nitride layer **14***b* are beryllium nitride layers including a nitrogen concentration gradient region, that is, in the case of forming a nitrogen concentration gradient region in the EUV transmissive membrane **10** with a three-layer structure of beryllium nitride/beryllium/beryllium nitride, when depositing the beryllium nitride membrane and metallic beryllium, the introduction of nitrogen gas may be stopped and switched to metallic beryllium deposition while continuing sputtering using a pure Be target with nitrogen gas in the chamber. In this way, a region is formed in which the nitrogen concentration in the film-deposited membrane decreases in the thickness direction as the concentration of nitrogen gas in the chamber decreases. On the other hand, in the case of switching the metallic beryllium to the beryllium nitride, the nitrogen concentration gradient region can be formed by starting the introduction of nitrogen gas in the middle of the process while sputtering is performed, contrary to the above. The thickness of the nitrogen concentration gradient region can be controlled by adjusting the time for which the nitrogen gas concentration is changed.

(3) Free-Standing Membrane Formation

[0052] An unnecessary portion of the Si substrate other than the outer edge of the Si substrate **28** where the composite membrane is formed, which is left as a border, is removed by etching to make the composite membrane free-standing. Etching of Si may be performed by any method, but etching with XeF.sub.2 is preferred. Subsequently, the exposed portions of the first protective layer **16***a* and the second protective layer **16***b* are removed to obtain the EUV transmissive membrane **10** with a three-layer configuration consisting of the metallic beryllium layer **12**, the first nitride layer **14***a*, and the second nitride layer **14***b* in the form of the pellicle **11**. Particularly, after the free-standing membrane forming step using a fluorine-based etching agent such as XeF.sub.2 described above, the first protective layer **16***a* and the second protective layer **16***b* become an unnecessary membrane in terms of reducing the EUV transmittance because the strong protective function against the fluorine-based etching agent is no longer required. Therefore, after the composite membrane is made into a free-standing membrane, the EUV transmittance can be further enhanced by removing the protective layers **16***a* and **16***b*. The processing for removing the amorphous carbon layer, which serves as the first protective layer **16***a* and the second protective layer **16***b*, is preferably performed by installing a composite membrane including the EUV transmissive membrane **10** into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated, then bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the composite membrane. The amorphous carbon layer, which serves as the first protective layer **16***a* and the second protective layer **16***b*, is exposed to a hydrogen plasma and/or hydrogen radical atmosphere or an oxygen plasma and/or oxygen radical atmosphere, whereby C on the surface of the amorphous carbon layer reacts with H or O to remove the amorphous carbon layer. As a result, the entire first protective layer **16***a* and the exposed portion of the second protective layer **16***b* can be eliminated. On the other hand, the second protective layer **16***b* interposed between the Si substrate **28** and the second nitride layer **14***b* is left intact to constitute the amorphous carbon layer **16** of the pellicle **11**.

Pellicle

[0053] Accordingly, as illustrated in FIG. **2**B(*k*), the EUV transmissive membrane **10** is preferably provided in the form of the pellicle **11**. The pellicle **11** includes the substrate **20**, the amorphous carbon layer **16**, and the EUV transmissive membrane **10**. The substrate **20** has a first side **20***a*, a second side **20***b*, and a cavity **26** at the center thereof. The substrate **20** is preferably a Si substrate or a border made of Si. The first side **20***a* of the substrate **20** is covered with the amorphous carbon layer **16**. The EUV transmissive membrane **10** covers a surface of the amorphous carbon layer **16** opposite to the substrate **20**, and the EUV transmissive membrane **10** is exposed as a free-standing membrane constituting a bottom surface of the cavity **26** at the same level as the surface of the amorphous carbon layer **16**.

[0054] The amorphous carbon layer **16** is a layer containing amorphous carbon. The amorphous carbon layer **16** located directly below the second nitride layer **14***b* (i.e., second protective layer **16***b*) contributes to the improvement of the strength of the second nitride layer **14***b*. In other words, the amorphous carbon layer **16** contributes to the improvement of crystallinity of the second nitride layer **14***b* deposited thereon, and as a result, the density and strength of the second nitride layer **14***b* can be enhanced. That is, when the second nitride layer **14***b* is directly deposited on the substrate **20** in the absence of the amorphous carbon layer **16**, the crystallinity of the nitride at the initial stage of deposition deteriorates, which leads to a decrease in the density and strength of the nitride. In this regard, when layer-deposited on the substrate **20** via the amorphous carbon layer **16**, a membrane with high nitride crystallinity is obtained as the second nitride layer **14***b*. Since the amorphous carbon layer **16** corresponds to the second protective layer **16***b* described above, the foregoing description of the second protective layer **16***b* is simply applied to the amorphous carbon layer **16**. Accordingly, the thickness of the amorphous carbon layer **16** is preferably 1 to 15 nm, more preferably 2 to 12 nm, and even more preferably 3 to 10 nm.

[0055] The pellicle **11** may further include a mask layer **22***a* that covers a second side **20***b* of the substrate **20**. The mask layer **22***a* is preferably a SiO.sub.2 layer.

Exposure Method

[0056] According to a preferred aspect of the present invention, the following exposure method using the EUV transmissive membrane **10** is provided. In this method, first, a composite membrane with a five-layer configuration including a first protective layer **16***a* and a second protective layer **16***b* that contain amorphous carbon on both sides of the EUV transmissive membrane **10** is prepared, as illustrated in FIG. **2**B(*j*). After the composite membrane with a five-layer configuration installed into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated, hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are brought into contact with the composite membrane with a five-layer configuration to remove exposed portions of the first protective layer **16***a* and the second protective layer **16***b*. At this time, as illustrated in FIG. **2**B(*k*), the second protective layer **16***b* interposed between the border **20** made of Si and the second nitride layer **14***b* is left intact to constitute the amorphous carbon layer **16** of the pellicle **11**. The pellicle **11** including the EUV transmissive membrane **10** with a three-layer configuration, from which the first protective layer **16***a* and the second protective layer **16***b* have been removed in this way, is installed into an EUV exposure apparatus, and then EUV is transmitted through the EUV transmissive membrane **10** to perform pattern exposure on a photosensitive substrate plate in the EUV exposure apparatus.

EXAMPLES

[0057] The present invention will be described in more detail with reference to the following examples. However, the present invention is not limited to the following examples.

Example 1 (Reference)

[0058] According to the procedures illustrated in FIGS. **2**A and **2**B, a composite free-standing membrane (EUV transmissive membrane **10** with protective layers **16***a* and **16***b* provided on both sides thereof) with a five-layer configuration of amorphous carbon/silicon nitride/Be/silicon nitride/amorphous carbon was fabricated as follows.

(1) Preparation of Si Substrate

[0059] A Si wafer **20** having a diameter of 8 inches (20.32 cm) was prepared (FIG. **2**A(*a*)). A SiO.sub.2 membrane **22** having a thickness of 50 nm was formed on both sides of the Si wafer **20** by thermal oxidation (FIG. **2**A(*b*)). Resist was applied to both sides of the Si wafer **20**, and a resist mask **24** for SiO.sub.2 etching was formed by exposure and development so that a 110 mm×145

mm resist hole was created on one side (FIG. **2**A(*c*)). An exposed portion of the SiO.sub.2 membrane **22** was etched and removed by wet-etching one side of the substrate with hydrofluoric acid to fabricate a SiO.sub.2 mask as the mask layer **22***a* (FIG. **2**A(*d*)). The resist mask **24** for SiO.sub.2 etching was removed with an ashing apparatus (FIG. **2**A(*e*)). Si was then wet-etched with a TMAH solution. This etching was performed only for an etching time to obtain a target Si substrate having thickness of **50** μm with an etching rate measured in advance (FIG. **2**A(*f*)). Finally, the SiO.sub.2 membrane **22** formed on the surface not subjected to Si etching was removed and cleaned with hydrofluoric acid to prepare a Si substrate **28** (FIG. **2**B(*g*)). The Si substrate outline may be diced with a laser **30**, if necessary (FIG. **2**B(*h*)), to achieve the desired shape (FIG. **2**B(*i*)). In this way, a 110 mm×145 mm cavity **26** was provided at the center of the 8-inch (20.32 cm) Si wafer **20** to prepare the Si substrate **28** having a Si thickness of 50 μm in the cavity **26** portion.
(2) Formation of Composite Membrane
[0060] On the Si substrate **28** including the cavity **26** obtained in (1) above, a composite membrane with a five-layer structure of amorphous carbon/silicon nitride/Be/silicon nitride/amorphous carbon was formed as follows (FIG. **2**B(*i*)). First, the Si substrate plate **28** was set in a multi-target sputtering apparatus, and a graphite target, Si.sub.3N.sub.4 target and a pure Be target were attached thereto. A chamber was evacuated, the graphite target was used to carry out sputtering only with argon gas at an internal pressure of 0.3 Pa, and the sputtering was terminated at the time when 2 nm of diamond-like carbon (DLC) was layer-deposited as amorphous carbon. Subsequently, the chamber was evacuated again, the Si.sub.3N.sub.4 target was used to carry out sputtering only with argon gas at an internal pressure of 0.7 Pa, and the sputtering was terminated at the time when 2 nm of Si.sub.3N.sub.4 was layer-deposited. Furthermore, the chamber was evacuated again, the pure Be target was used to carry out sputtering only with argon gas at an internal pressure of 0.5 Pa, and the sputtering was terminated at the time when 20 nm of beryllium was layer-deposited. Subsequently, the chamber was evacuated again, the Si.sub.3N.sub.4 target was used to carry out sputtering only with argon gas at an internal pressure of 0.7 Pa, and the sputtering was terminated at the time when 2 nm of Si.sub.3N.sub.4-x was layer-deposited. Thereafter, sputtering was performed using the graphite target in the same manner as in the first step, and the sputtering was terminated at the time when 2 nm of amorphous carbon was layer-deposited. In this manner, a composite membrane with 2 nm of amorphous carbon (C)/2 nm of silicon nitride (Si.sub.3N.sub.4-x)/20 nm of beryllium (Be)/2 nm of silicon nitride (Si.sub.3N.sub.4-x)/2 nm of amorphous carbon (C) was formed. In other words, the composite membrane has a five-layer configuration consisting of the EUV transmissive membrane **10**, which is composed of three layers of the first nitride layer **14***a*, the metallic beryllium layer **12**, and the second nitride layer **14***b*, and the first protective layer **16***a*, and the second protective layer **16***b*, wherein the first protective layer **16***a* and the second protective layer **16***b* containing amorphous carbon as a main component are formed on both sides of the EUV transmissive membrane **10**.
(3) Free-Standing Membrane Formation
[0061] The Si substrate **28** with the composite membrane prepared in (2) above was set in a chamber of an XeF.sub.2 etcher capable of processing an 8-inch (20.32 cm) substrate. The chamber was sufficiently evacuated. At this time, if moisture remains in the chamber, the moisture reacts with the XeF.sub.2 gas to generate hydrofluoric acid, and corrosion of the etcher or unexpected etching occurs. Therefore, the sufficient evacuation was performed. If necessary, vacuuming and nitrogen gas introduction were repeated in the chamber to reduce residual moisture in the chamber. When the sufficient evacuation was achieved, a valve between a XeF.sub.2 material tank and a preliminary space was opened. As a result, XeF.sub.2 was sublimated, and XeF.sub.2 gas was also accumulated in the preliminary space. When the XeF.sub.2 gas was sufficiently accumulated in the preliminary space, the valve between the preliminary space and the chamber was opened to introduce the XeF.sub.2 gas into the chamber. The XeF.sub.2 gas was decomposed into Xe and F, and F reacted with Si to generate SiF.sub.4. Since the boiling point of SiF.sub.4 was −95° C.,

SiF.sub.4 generated was rapidly evaporated, causing a reaction of F with the newly exposed Si substrate. When the Si etching proceeded and F in the chamber decreased, the chamber was evacuated, and the XeF.sub.2 gas was introduced into the chamber again to perform the etching. In this manner, the evacuation, the introduction of the XeF.sub.2 gas, and the etching were repeated, and the etching was continued until the Si substrate **28** corresponding to the portion to be formed into the free-standing membrane disappeared. When the Si substrate of the unnecessary portion disappeared, the etching was terminated. In this way, a composite free-standing membrane with a five-layer configuration having a border **20** made of Si is obtained (FIG. **2**B(*j*)).

Example 2 (Reference)

[0062] A composite free-standing membrane (EUV transmissive membrane **10** with protective layers **16***a* and **16***b* provided on both sides thereof) having a border **20** made of Si was fabricated in the same manner as in Example 1. Thereafter, amorphous carbon layers exposed on both sides of the composite free-standing membrane were etched with hydrogen plasma to reduce the thickness of each amorphous carbon layer to 1 nm. In this manner, a composite free-standing membrane with a five-layer configuration consisting of 1 nm of amorphous carbon (C)/2 nm of silicon nitride (Si.sub.3N.sub.4-x)/20 nm of beryllium (Be)/2 nm of silicon nitride (Si.sub.3N.sub.4-x)/1 nm of amorphous carbon (C), which had a border **20** made of Si, was obtained (FIG. **2**B(*j*)).

Example 3

[0063] A composite free-standing membrane (EUV transmissive membrane **10** with protective layers **16***a* and **16***b* provided on both sides thereof) having a border **20** made of Si was fabricated in the same manner as in Example 1. Thereafter, amorphous carbon layers exposed on both sides of the EUV transmissive membrane **10** were etched with hydrogen plasma to remove each of the amorphous carbon layers other than the amorphous carbon layer **16** in a portion sandwiched between the EUV transmissive membrane **10** and the border **20** made of Si. In this manner, a composite free-standing membrane (i.e., EUV transmissive membrane **10**) with a three-layer configuration consisting of 2 nm of silicon nitride (Si.sub.3N.sub.4-x)/20 nm of beryllium (Be)/2 nm of silicon nitride (Si.sub.3N.sub.4-x), which had the border **20** made of Si, was obtained in the form of the pellicle **11** (FIG. **2**B(*k*)).

Example 4

[0064] A composite free-standing membrane (EUV transmissive membrane **10** with protective layers **16***a* and **16***b* provided on both sides thereof) having a border **20** made of Si was fabricated in the same manner as in Example 1, except that a composite membrane with 12 nm of amorphous carbon (C)/1 nm of silicon nitride (Si.sub.3N.sub.4-x)/20 nm of beryllium (Be)/1 nm of silicon nitride (Si.sub.3N.sub.4-x)/12 nm of amorphous carbon (C) was formed by changing each of the deposition time of amorphous carbon and silicon nitride in (2) above. Thereafter, amorphous carbon layers exposed on both sides of the EUV transmissive membrane **10** were etched with hydrogen plasma to remove each of the amorphous carbon layers other than the amorphous carbon layer **16** in a portion sandwiched between the EUV transmissive membrane **10** and the border **20** made of Si. In this manner, a composite free-standing membrane (i.e., EUV transmissive membrane **10**) with a three-layer configuration consisting of 1 nm of silicon nitride (Si.sub.3N.sub.4-x)/20 nm of beryllium (Be)/1 nm of silicon nitride (Si.sub.3N.sub.4-x), which had the border **20** made of Si, was obtained in the form of the pellicle **11** (FIG. **2**B(*k*)).

EUV Transmittance and in-Plane Uniformity Thereof

[0065] EUV light was irradiated onto a free-standing membrane as the EUV transmissive membrane fabricated in Examples 1 to 3 at a power of 600 W in a hydrogen atmosphere of 20 Pa for 15 minutes, and then the amount of transmitted EUV light was measured with a sensor. When the EUV transmittance was determined by comparing the obtained measurement value with a value obtained by directly measuring the amount of EUV light with a sensor without using the EUV transmissive membrane, the results shown in Table 1 were obtained. At this time, a spot of the EUV light used for measuring the transmittance had an elliptical shape of 0.5 mm×0.2 mm, and the EUV

transmittance was measured at a wavelength of 13.5 nm within the spot size. Therefore, in order to evaluate the in-plane uniformity of the EUV transmittance, the EUV transmittance at each place was determined by measuring the EUV transmittance while moving the spot of the EUV light, and the value of the in-plane variation was calculated from the data as a value three times the standard deviation. Accordingly, the smaller value of the in-plane variation refers to better in-plane uniformity of the EUV transmittance.
TABLE-US-00001 TABLE 1 EUV properties Component and thickness (nm)of each layer In-plane First First Metallic Second Second variation in protective nitride beryllium nitride protective EUV EUV layer layer layer layer layer transmittance transmittance C Si.sub.3N.sub.4−x Be Si.sub.3N.sub.4−x C (%) (%) Example 2 2 20 2 2 91.1 0.2 1* Example 1 2 20 2 1 92.6 0.3 2* Example None 2 20 2 None 93.8 0.5 3 Example None 1 20 1 None 94.7 0.4 4 *represents a reference example.
[0066] As can be seen from the results, when the amorphous carbon layer is left as in Examples 1 and 2, the in-plane variation in the EUV transmittance is small. Thus, while the in-plane uniformity becomes excellent, the EUV transmittance becomes low. In contrast, the removal of the amorphous carbon layer by etching, as in Examples 3 and 4, can increase a transmittance while ensuring acceptable in-plane uniformity. Although a smaller in-plane variation in the EUV transmittance has the advantage of making the in-plane exposure amount more uniform and improving device homogeneity, a higher EUV transmittance results in shorter exposure time, that is, higher throughput is achieved. In other words, the present invention provides an EUV transmissive membrane that exhibits high EUV transmittance.
[0067] In addition, Example 4 is an aspect in which the thickness of the amorphous carbon layer as the first protective layer **16***a* and the second protective layer **16***b* is 12 μm, which is greater than the thickness (2 μm) of the amorphous carbon layer of Examples 1 to 3, whereby the protective function against a fluorine-based etching agent is enhanced. As shown in Table 1, it is possible to realize high EUV transmittance by finally removing the protective layers **16***a* and **16***b* in the present invention, so that the protective function described above can be enhanced by thickening the amorphous carbon layer as the protective layers **16***a* and **16***b* as in Example 4.

Claims

1. An EUV transmissive membrane with a three-layer configuration consisting of: a metallic beryllium layer having a first side and a second side, a first nitride layer that covers the first side of the beryllium layer, wherein the first nitride layer comprises at least one selected from the group consisting of silicon nitride, boron nitride, and zirconium nitride, and a second nitride layer that covers the second side of the beryllium layer, wherein the second nitride layer comprises at least one selected from the group consisting of silicon nitride, boron nitride, and zirconium nitride, wherein the EUV transmissive membrane has an EUV transmittance of 88% or more at a wavelength of 13.5 nm.
2. The EUV transmissive membrane according to claim 1, wherein the EUV transmissive membrane has a thickness of 7 to 30 nm.
3. The EUV transmissive membrane according to claim 1, wherein the beryllium layer has a thickness of 5 to 25 nm.
4. The EUV transmissive membrane according to claim 1, wherein the first nitride layer and the second nitride layer each have a thickness of 1 to 5 nm.
5. A pellicle comprising: a substrate having a first side and a second side, and a cavity at a center of the substrate, an amorphous carbon layer that covers the first side of the substrate, and the EUV transmissive membrane according to claim 1, which covers a surface of the amorphous carbon layer opposite to the substrate, wherein the EUV transmissive membrane is exposed as a free-standing membrane constituting a bottom surface of the cavity at a same level as a surface of the

amorphous carbon layer.

6. The pellicle according to claim 5, wherein the amorphous carbon layer has a thickness of 1 to 15 nm.

7. The pellicle according to claim 5, wherein the substrate is a Si substrate.

8. The pellicle according to claim 5, further comprising a mask layer that covers the second side of the substrate.

9. The pellicle according to claim 8, wherein the mask layer is a SiO.sub.2 layer.

10. An exposure method, comprising the steps of: providing a composite membrane with a five-layer configuration, wherein the composite membrane comprises a protective layer containing amorphous carbon on both sides of the EUV transmissive membrane according to claim 1; installing the composite membrane with a five-layer configuration into an apparatus in which hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals are generated; bringing hydrogen plasma and/or hydrogen radicals or oxygen plasma and/or oxygen radicals into contact with the composite membrane with a five-layer configuration to remove the protective layer; and installing the EUV transmissive membrane with a three-layer configuration, from which the protective layer has been removed, into an EUV exposure apparatus, then transmitting EUV through the EUV transmissive membrane to perform pattern exposure on a photosensitive substrate plate in the EUV exposure apparatus.